# Mathematical Connectivity of Crystallographic Phase Map Paradigms

A Strict Derivation Connecting Ground-Truth Density,
Experimentally Phased Maps (SAD/MAD),
Classical Estimators, and the EDENN Stochastic Framework

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*This document establishes the precise mathematical relationship between the theoretical ground-truth electron density, experimentally phased maps from anomalous dispersion (SAD/MAD), and the family of model-based phase estimators: Blow-Crick, Sigma-A (Read 1986), Lunin–Skovoroda LB estimates, composite OMIT maps, PanDDA, and the EDENN stochastic omit ensemble. Every map is expressed as a linear functional of a single underlying object — the complex structure-factor field $F(\mathbf{h})$ — and the connectivity between paradigms is expressed as a diagram of commuting operators in Hilbert space.*

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# §1 The Theoretical Ground-Truth Map

## 1.1 Electron density as a square-integrable function

Let the unit cell occupy volume $V \subset \mathbb{R}^3$. The electron density $\rho : V \to \mathbb{R}_{\geq 0}$ is a non-negative, square-integrable, real-valued function. In quantum mechanics it equals the expectation value of the electron number-density operator over all electronic states.

**Definition 1.1. True electron density**

The ground-truth electron density is:

$$\rho_{\text{true}}(\mathbf{r}) = \sum_{j=1}^{N_{\text{atoms}}} \rho_j(\mathbf{r} - \mathbf{r}_j) \tag{1}$$

where $\rho_j$ is the scattering density of atom $j$ (the Fourier transform of its atomic scattering factor $f_j(s)$ convolved with a Debye–Waller envelope) and $\mathbf{r}_j$ is its equilibrium position. In the harmonic approximation:

$$\rho_j(\mathbf{r}) = f_j * \mathcal{N}(0, \mathbf{U}_j)(\mathbf{r}) \tag{2}$$

where $\mathbf{U}_j$ is the anisotropic displacement parameter tensor.

## 1.2 The Fourier representation

Because the crystal is periodic with unit-cell lattice vectors $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$, the density has a convergent Fourier series:

$$\rho_{\text{true}}(\mathbf{r}) = \frac{1}{V} \sum_{\mathbf{h} \in \mathbb{Z}^3} F_{\text{true}}(\mathbf{h})\, e^{-2\pi i\, \mathbf{h}\cdot\mathbf{r}} \tag{3}$$

with complex Fourier coefficients:

$$F_{\text{true}}(\mathbf{h}) = \int_V \rho_{\text{true}}(\mathbf{r})\, e^{2\pi i\, \mathbf{h}\cdot\mathbf{r}}\, d\mathbf{r} = |F_{\text{true}}(\mathbf{h})|\, e^{i\varphi_{\text{true}}(\mathbf{h})} \tag{4}$$

The modulus $|F_{\text{true}}(\mathbf{h})|$ and phase $\varphi_{\text{true}}(\mathbf{h}) \in (-\pi, \pi]$ are both real numbers that depend on the full atomic arrangement.

**Definition 1.2. Ground-truth map operator**

Define the **ground-truth map operator** $\mathcal{T} : \mathbb{C}^{|\mathcal{H}|} \to L^2(V)$ by:

$$(\mathcal{T}\,\mathbf{F})(\mathbf{r}) = \frac{1}{V} \sum_{\mathbf{h}} F(\mathbf{h})\, e^{-2\pi i\, \mathbf{h}\cdot\mathbf{r}} = \mathcal{F}^{-1}[\mathbf{F}](\mathbf{r}) \tag{5}$$

The ground-truth density is then $\rho_{\text{true}} = \mathcal{T}[F_{\text{true}}(\cdot)]$.

## 1.3 What the diffraction experiment measures

A monochromatic X-ray diffraction experiment measures intensities:

$$I(\mathbf{h}) = |F_{\text{true}}(\mathbf{h})|^2 + \varepsilon(\mathbf{h}) \tag{6}$$

where $\varepsilon(\mathbf{h}) \sim \mathcal{N}(0, \sigma_I^2(\mathbf{h}))$ is measurement noise. After Lorentz–polarisation corrections and scaling, this yields amplitudes:

$$|F_{\text{obs}}(\mathbf{h})| = \sqrt{I(\mathbf{h})} \approx |F_{\text{true}}(\mathbf{h})| \quad (\text{if } \varepsilon \text{ small}) \tag{7}$$

**The phases $\varphi_{\text{true}}(\mathbf{h})$ are entirely absent from equation (6).** This is the crystallographic phase problem — an irreducible consequence of measuring probability (intensity) rather than amplitude.

*Remark 1.1. Completeness of the amplitude information*

The mapping $\rho_{\text{true}} \mapsto \{|F_{\text{true}}(\mathbf{h})|\}_{\mathbf{h}}$ is many-to-one: infinitely many density functions are consistent with any given set of measured amplitudes. Determining $\rho_{\text{true}}$ from amplitudes alone is an ill-posed inverse problem. Phase estimation is the art of selecting a unique solution using additional constraints (atomicity, solvent flatness, non-negativity, chemical prior knowledge).

## §2 Experimentally Phased Maps: SAD and MAD

### 2.1 Do SAD/MAD maps contain the true phases?

This is a precisely answerable question, and the answer has a surprising nuance: **SAD/MAD phasing provides experimental information that constrains $\varphi_{\text{true}}(\mathbf{h})$, but the resulting phases are not equal to $\varphi_{\text{true}}(\mathbf{h})$.** They are Bayesian posterior estimates of $\varphi_{\text{true}}$ from anomalous signal, and they carry irreducible uncertainty.

### 2.2 The anomalous dispersion signal

For an atom with anomalous scatterers (e.g. Se-Met for SAD, multiple wavelengths for MAD), the atomic scattering factor takes the form:

$$f = f^0 + f' + if'' \tag{8}$$

where $f^0$ is the normal elastic term, $f' = \Delta f'$ is the real anomalous correction, and $f'' = \Delta f''$ is the imaginary correction. The imaginary term $f''$ breaks the Friedel symmetry, producing measurable differences between Friedel mates:

$$\Delta I^{\pm}(\mathbf{h}) = I(\mathbf{h}) - I(-\mathbf{h}) \neq 0 \tag{9}$$

### 2.3 Exact algebraic structure of the SAD phase constraint

For a structure with heavy atoms at positions $\{\mathbf{s}_j\}$, the total structure factor decomposes as:

$$F_{\text{true}}(\mathbf{h}) = F_P(\mathbf{h}) + F_A(\mathbf{h}) \tag{10}$$

where $F_P$ is the contribution from normal (non-anomalous) atoms and $F_A = F_A^0 + F_A' + iF_A''$ is the anomalous contribution. The Friedel difference is:

$$\Delta F^{\pm}(\mathbf{h}) = F(\mathbf{h}) - F(-\mathbf{h})^* = 2i\, F_A''(\mathbf{h}) \tag{11}$$

The measurable Bijvoet difference amplitude is:

$$|\Delta F_{\text{anom}}(\mathbf{h})| = \frac{|\Delta F^{\pm}|}{2} = |F_A''(\mathbf{h})| \tag{12}$$

### 2.4 The Harker construction and phase ambiguity

From $|F_{\text{obs}}(\mathbf{h})|$ and $|\Delta F_{\text{anom}}(\mathbf{h})|$, the Harker construction shows that $\varphi_{\text{true}}(\mathbf{h})$ is constrained to **exactly two solutions** lying on a circle in the Argand plane — the phase ambiguity:

$$\varphi_{\text{true}}(\mathbf{h}) \in \left\{\varphi_{\text{SAD}}(\mathbf{h}) + \delta(\mathbf{h}),\ \varphi_{\text{SAD}}(\mathbf{h}) - \delta(\mathbf{h})\right\} \tag{13}$$

where $\varphi_{\mathrm{SAD}}(\mathbf{h})$ is the phase of $F''_A(\mathbf{h})$ (known from the anomalous substructure solution) and:

$$\delta(\mathbf{h}) = \arccos\left(\frac{|F_{\mathrm{obs}}(\mathbf{h})|^2 + |F''_A(\mathbf{h})|^2 - |F_P(\mathbf{h})|^2}{2\,|F_{\mathrm{obs}}(\mathbf{h})|\,|F''_A(\mathbf{h})|}\right) \tag{14}$$

This is **not** the true phase — it is a geometric constraint with two solutions.

### 2.5 The SAD phase probability distribution

The phase ambiguity is resolved probabilistically. The likelihood of the observed anomalous difference $|\Delta F_{\mathrm{anom}}|$ given phase $\varphi_h$ is:

$$\mathcal{L}(|\Delta F_{\mathrm{anom}}| \mid \varphi_h) \propto \exp\left[-\frac{\left(|\Delta F_{\mathrm{anom}}| - |F''_A|\cos(\varphi_h - \varphi_{\mathrm{SAD}})\right)^2}{2\,\sigma^2_{\Delta F_{\mathrm{anom}}}}\right] \tag{15}$$

Combined with the native amplitude likelihood (Rice distribution from Wilson statistics), the posterior phase probability is:

$$P_{\mathrm{SAD}}(\varphi_h) \propto \exp\Big[A(\mathbf{h})\cos\varphi_h + B(\mathbf{h})\sin\varphi_h\Big] \tag{16}$$

which is a **generalised von Mises distribution** (Hendrickson 1979). The coefficients $A$, $B$ are functions of the Bijvoet differences and their errors.

**Theorem 2.1. SAD phase distribution is von Mises**

The SAD posterior phase distribution (16) is equivalent to the von Mises distribution $\mathrm{vonMises}(\varphi^{\mathrm{SAD}}_{\mathrm{mod}}, X^{\mathrm{SAD}}_h)$ with:

$$\varphi^{\mathrm{SAD}}_{\mathrm{mod}}(\mathbf{h}) = \arctan(B(\mathbf{h})/A(\mathbf{h})) \tag{17}$$

$$X^{\mathrm{SAD}}_h = \sqrt{A(\mathbf{h})^2 + B(\mathbf{h})^2} \tag{18}$$

*Proof.* Write $A\cos\varphi_h + B\sin\varphi_h = \sqrt{A^2+B^2}\cos(\varphi_h - \arctan(B/A)) = X^{\mathrm{SAD}}_h\cos(\varphi_h - \varphi^{\mathrm{SAD}}_{\mathrm{mod}})$. Substituting into (16) gives the von Mises kernel with these parameters. The normalisation constant $1/(2\pi I_0(X^{\mathrm{SAD}}_h))$ follows from the standard von Mises integral. □ □

*Remark 2.1. What SAD phases actually are*

The phases output by SHELXE, PHENIX, CCP4 after SAD phasing are the *centroids* of $P_{\mathrm{SAD}}(\varphi_h)$, i.e. $\varphi^{\mathrm{SAD}}_{\mathrm{mod}}$ from (17), with associated FOMs $m^{\mathrm{SAD}}_h = I_1(X^{\mathrm{SAD}}_h)/I_0(X^{\mathrm{SAD}}_h)$. They are **posterior mean phases**, not true phases. The difference $|\varphi^{\mathrm{SAD}}_{\mathrm{mod}} - \varphi_{\mathrm{true}}|$ is the residual phase error after SAD phasing, which can be several tens of degrees at lower resolution and shrinks (but never reaches zero) with improved anomalous signal. SAD phases are the *best currently available estimate* of the true phase given anomalous diffraction data — they are not the truth itself.

### 2.6 MAD phasing: multi-wavelength constraint

MAD phasing measures $\{I(\mathbf{h},\lambda_i)\}_{i=1}^{n_\lambda}$ at $n_\lambda \geq 3$ wavelengths near an absorption edge. At each wavelength:

$$F(\mathbf{h},\lambda_i) = F_P(\mathbf{h}) + f'(\lambda_i)F^0_A(\mathbf{h}) + if''(\lambda_i)F^0_A(\mathbf{h}) \tag{19}$$

The system of equations for $n_\lambda$ wavelengths overdetermines the unknowns $|F_P|$, $|F^0_A|$, and $\varphi_{\mathrm{true}} - \varphi_A$. The phase ambiguity of SAD is resolved to a unique solution in the ideal case:

$$P_{\mathrm{MAD}}(\varphi_h) \approx \delta(\varphi_h - \varphi_{\mathrm{true}}(\mathbf{h})) \quad \text{as } n_\lambda \to \infty,\ \sigma_I \to 0 \tag{20}$$

In practice with finite noise and $n_\lambda = 3$–4, MAD gives a sharper posterior than SAD (larger $X_h^{\mathrm{MAD}}$) but the distribution remains non-degenerate.

*Remark 2.2. MAD phases and truth*

MAD phases are the closest experimentally accessible approximation to $\varphi_{\mathrm{true}}$. In ideal conditions at atomic resolution with perfect data, the MAD phase posterior approaches a delta function. In practice, residual errors of 15–25° are typical at 2 Å resolution. Even MAD-phased maps do not contain the "true" phases — they contain experimentally constrained probabilistic estimates thereof.

### 2.7 Connection to the EDENN framework

SAD/MAD phases provide the closest available ground truth for validating any phase estimator. The connection to EDENN is via three channels:

1. **FOM validation:** $m_h^{\mathrm{SAD}}$ provides an independent estimate of the concentration parameter $X_h$. If EDENN's $\hat{X}_h$ is calibrated, it should agree with $X_h^{\mathrm{SAD}}$ in each resolution shell.
2. **Phase error ground truth:** $\varphi_{\mathrm{mod}}^{\mathrm{SAD}}$ can substitute for $\varphi_{\mathrm{true}}$ when computing empirical phase errors for model phases, enabling the key benchmark comparison.
3. **Map comparison:** the SAD-phased density $\rho_{\mathrm{SAD}} = \mathcal{F}^{-1}[m_h^{\mathrm{SAD}}|F_{\mathrm{obs}}|e^{i\varphi_{\mathrm{mod}}^{\mathrm{SAD}}}]$ is the best single-experiment approximation to $\rho_{\mathrm{true}}$, and provides the reference map against which all model-based estimators are compared.

## §3 The Family of Map Estimators: A Unified Framework

### 3.1 The general map estimator

Every crystallographic map used in practice can be written in the form:

**Definition 3.1. General map estimator**

A **map estimator** is an element of the class:

$$\rho^{\mathcal{E}}(\mathbf{r}) = \frac{1}{V}\sum_{\mathbf{h}} w_h\,|F_{\mathrm{obs}}(\mathbf{h})|\,e^{i\hat{\varphi}_h}\,e^{-2\pi i\,\mathbf{h}\cdot\mathbf{r}} \tag{21}$$

parameterised by two functions on reciprocal space:

- $w_h \in [0,1]$: an amplitude weight (FOM or related quantity)
- $\hat{\varphi}_h \in (-\pi,\pi]$: a phase estimate

Different paradigms correspond to different choices of $(w_h, \hat{\varphi}_h)$.

All paradigms use observed amplitudes $|F_{\mathrm{obs}}(\mathbf{h})|$ in the numerator — this is non-negotiable because the true amplitudes $|F_{\mathrm{true}}(\mathbf{h})|$ are the closest observable proxy. The entire information content of a map is therefore encoded in the pair $(w_h, \hat{\varphi}_h)$.

### 3.2 The distance between two estimators

**Definition 3.2. Map distance**

For two map estimators $\rho^{\mathcal{A}}$ and $\rho^{\mathcal{B}}$ with parameters $(w_h^A, \hat{\varphi}_h^A)$ and $(w_h^B, \hat{\varphi}_h^B)$, the real-space $L^2$ distance is:

$$\|\rho^A - \rho^B\|_2^2 = \frac{1}{V^2} \sum_{\mathbf{h}} \left| w_h^A |F_{\text{obs}}| e^{i\hat{\varphi}_h^A} - w_h^B |F_{\text{obs}}| e^{i\hat{\varphi}_h^B} \right|^2 \tag{22}$$

by Parseval's theorem. The distance is zero if and only if $w_h^A = w_h^B$ and $\hat{\varphi}_h^A = \hat{\varphi}_h^B$ for all $\mathbf{h}$.

### 3.3 The ground-truth distance

**Definition 3.3. Mean-square map error**

The mean-square error (MSE) of estimator $\rho^{\mathcal{E}}$ relative to $\rho_{\text{true}}$ is:

$$\begin{aligned} \text{MSE}[\rho^{\mathcal{E}}] &= \mathbb{E}_{\varphi_h}\left[\|\rho^{\mathcal{E}} - \rho_{\text{true}}\|_2^2\right] \\ &= \frac{1}{V^2} \sum_{\mathbf{h}} \mathbb{E}_{\varphi_h}\left[ |w_h |F_{\text{obs}}| e^{i\hat{\varphi}_h} - |F_{\text{true}}| e^{i\varphi_{\text{true}}}|^2 \right] \end{aligned} \tag{23}$$

where the expectation is taken over the phase probability distribution $P(\varphi_h)$ representing our uncertainty.

## §4 Classical Blow–Crick: The MMSE Estimator

### 4.1 Derivation of the optimal weights

**Theorem 4.1. Blow–Crick optimality (strict derivation)**

Among all estimators of the form (21) with fixed phase estimate $\hat{\varphi}_h = \varphi_{\text{mod}}(\mathbf{h})$, the unique minimiser of the MSE (23) is:

$$w_h^* = m_h \equiv \mathbb{E}[e^{i(\varphi_h - \varphi_{\text{mod}})}] = \frac{I_1(X_h)}{I_0(X_h)} \tag{24}$$

giving the Blow–Crick best map:

$$\rho_{\text{BC}}(\mathbf{r}) = \frac{1}{V} \sum_{\mathbf{h}} m_h \, |F_{\text{obs}}(\mathbf{h})| \, e^{i\varphi_{\text{mod}}(\mathbf{h})} \, e^{-2\pi i \, \mathbf{h}\cdot\mathbf{r}} \tag{25}$$

*Proof.* Expand the MSE term by term. For reflection $\mathbf{h}$:

$$\begin{aligned} &\mathbb{E}[|w_h|F_{\text{obs}}|e^{i\varphi_{\text{mod}}} - |F_{\text{true}}|e^{i\varphi_{\text{true}}}|^2] && (26) \\ &= w_h^2|F_{\text{obs}}|^2 + |F_{\text{true}}|^2 - 2w_h|F_{\text{obs}}||F_{\text{true}}|\,\mathbb{E}[\cos(\varphi_{\text{true}} - \varphi_{\text{mod}})] && (27) \\ &\approx w_h^2|F_{\text{obs}}|^2 + |F_{\text{true}}|^2 - 2w_h|F_{\text{obs}}|^2\,\mathbb{E}[\cos(\varphi_h - \varphi_{\text{mod}})] && (28) \end{aligned}$$

where in the last step we use $|F_{\text{true}}| \approx |F_{\text{obs}}|$ (small measurement noise) and $\mathbb{E}[\cos(\varphi_{\text{true}} - \varphi_{\text{mod}})] = m_h$ by definition. Setting $\partial/\partial w_h = 0$:

$$2w_h|F_{\text{obs}}|^2 - 2|F_{\text{obs}}|^2 m_h = 0 \implies w_h = m_h \tag{29}$$

This is a global minimum (the second derivative $2|F_{\text{obs}}|^2 > 0$). Substituting $w_h = m_h$ into (21) gives (25). The Bessel ratio form $m_h = I_1(X_h)/I_0(X_h)$ follows from evaluating $\mathbb{E}[e^{i(\varphi_h - \varphi_{\text{mod}})}] = \int e^{i\theta} P(\theta) d\theta$ for the von Mises distribution, yielding the characteristic function at $n = 1$. □ □

### 4.2 The minimum achievable MSE

Substituting $w_h^* = m_h$ back into the MSE:

$$\text{MSE}[\rho_{\text{BC}}] = \frac{1}{V^2} \sum_{\mathbf{h}} |F_{\text{obs}}(\mathbf{h})|^2 (1 - m_h^2) \tag{30}$$

This is the **Cramér–Rao lower bound** for the class of linear estimators with phases fixed at $\varphi_{\text{mod}}$. Any other choice of $w_h$ gives a strictly larger MSE.

### 4.3 Connection to the true density

The Blow-Crick map relates to the true density via:

$$\rho_{\text{BC}}(\mathbf{r}) = \mathbb{E}_{\varphi_h}[\rho_{\text{true}}(\mathbf{r})] + \underbrace{\delta_{\text{amp}}(\mathbf{r}) + \delta_{\text{noise}}(\mathbf{r})}_{\text{small corrections}} \tag{31}$$

where $\delta_{\text{amp}}$ accounts for $|F_{\text{obs}}| \neq |F_{\text{true}}|$ (measurement noise in amplitudes) and $\delta_{\text{noise}}$ for finite completeness of the reflection set. In the ideal limit $|F_{\text{obs}}| = |F_{\text{true}}|$ and complete data:

$$\rho_{\text{BC}} = \mathbb{E}_{\varphi_h}[\rho_{\text{true}}] \tag{32}$$

**The Blow-Crick map is the expected true density averaged over the phase probability distribution.**

## §5 Sigma-A Estimation: Likelihood-Based FOM

### 5.1 The Read coordinate error model

Read (1986) derived the phase probability distribution from a statistical model of coordinate errors. For atoms with isotropic displacement $\sigma_x^2$, the expected attenuation of the model structure factor is:

$$\mathbb{E}[F_{\text{mod}}(\mathbf{h})] = D_h\, F_{\text{true}}(\mathbf{h}), \qquad D_h = \sigma_A(s) = \exp(-2\pi^2 \sigma_x^2 s^2) \tag{33}$$

The residual $\varepsilon(\mathbf{h}) = F_{\text{true}}(\mathbf{h}) - D_h F_{\text{mod}}(\mathbf{h})$ is modelled as isotropic complex Gaussian with variance $\Sigma_h$ per component. Under this model, the distribution of $F_{\text{obs}}$ given $F_{\text{mod}}$ is:

**Lemma 5.1. Rice distribution for acentric reflections**

For acentric reflections:

$$P(F_{\text{obs}} \mid F_{\text{mod}}, D_h, \Sigma_h) = \frac{2|F_{\text{obs}}|}{\Sigma_h} \exp\left(-\frac{|F_{\text{obs}}|^2 + D_h^2 |F_{\text{mod}}|^2}{\Sigma_h}\right) I_0\left(\frac{2D_h |F_{\text{obs}}||F_{\text{mod}}|}{\Sigma_h}\right) \tag{34}$$

For centric reflections the analogous distribution is the folded normal.

### 5.2 Phase probability from the Rice likelihood

Integrating the full complex structure-factor likelihood over $|F_{\text{obs}}|$ and extracting the phase component:

$$P_{\Sigma A}(\varphi_h \mid F_{\text{obs}}, F_{\text{mod}}) \propto \exp\left[X_h^{\Sigma A} \cos(\varphi_h - \varphi_{\text{mod}})\right] \tag{35}$$

with concentration:

$$X_h^{\Sigma A} = \frac{2\, D_h\, |F_{\text{obs}}(\mathbf{h})|\, |F_{\text{mod}}(\mathbf{h})|}{\Sigma_h} \tag{36}$$

The Sigma-A FOM is therefore:

$$m_h^{\Sigma A} = \frac{I_1(X_h^{\Sigma A})}{I_0(X_h^{\Sigma A})} \tag{37}$$

### 5.3 Working-set estimation and its bias

Parameters $(D_h, \Sigma_h)$ are estimated by maximising the product likelihood over all working-set reflections in resolution shell $k$:

$$(\hat{D}_{hk}, \hat{\Sigma}_{hk}) = \arg\max_{D,\Sigma} \prod_{\mathbf{h}\in\mathcal{W}_k} P(|F_{\text{obs}}(\mathbf{h})| \mid |F_{\text{mod}}(\mathbf{h})|, D, \Sigma) \tag{38}$$

**Proposition 5.1. Working-set inflation**

If the model was refined against $\mathcal{W}$, then the working-set likelihood estimator satisfies:

$$\mathbb{E}_{\mathcal{W}}[\hat{D}_{hk}] \geq D_k^{\text{true}}, \qquad \mathbb{E}_{\mathcal{W}}[\hat{\Sigma}_{hk}] \leq \Sigma_h^{\text{true}} \tag{39}$$

causing $\hat{m}_h^{\Sigma A} \geq m_h^{\text{true}}$.

*Argument.* Refinement against $\mathcal{W}$ minimises a target $\mathcal{T}(F_{\text{mod}} \mid \{|F_{\text{obs}}|\}_{\mathbf{h}\in\mathcal{W}})$. This is equivalent to maximising the working-set likelihood, so the refined model parameters already represent the global maximum of the likelihood for the working set. Estimating $(\hat{D}, \hat{\Sigma})$ from the same data therefore encounters a statistically over-determined system: the model phases $\varphi_{\text{mod}}$ are the phases that maximise $\prod_{\mathcal{W}} P(|F_{\text{obs}}| \mid F_{\text{mod}})$, so the ML estimate of $D$ is systematically biased upward and that of $\Sigma_h$ downward. In the extreme case of an exactly fitted model, $\hat{\Sigma}_{hk} \to 0$ and $\hat{m}_h^{\Sigma A} \to 1$ regardless of true phase accuracy. □ □

### 5.4 Connecting Sigma-A map to Blow-Crick

The Sigma-A map is:

$$\rho_{\Sigma\text{A}}(\mathbf{r}) = \mathcal{F}^{-1}\left[m_h^{\Sigma A}\, |F_{\text{obs}}|\, e^{i\varphi_{\text{mod}}}\right] \tag{40}$$

This has exactly the Blow-Crick form (25) with $m_h$ replaced by the (potentially inflated) Sigma-A estimate $m_h^{\Sigma A}$.

$$\boxed{\rho_{\Sigma\text{A}} = \rho_{\text{BC}} + \underbrace{\mathcal{F}^{-1}\left[(m_h^{\Sigma A} - m_h^{\text{true}})\, |F_{\text{obs}}|\, e^{i\varphi_{\text{mod}}}\right]}_{\text{bias term}\geq 0}} \tag{41}$$

The bias term is non-negative on average (positive map values inflated by overestimated FOMs).

## §6 Lunin–Skovoroda LB Estimates: Unbiased Sigma-A

### 6.1 The R-free correction

Lunin & Skovoroda (1995) observed that replacing $\mathcal{W}$ with the R-free test set $\mathcal{F}$ in (38) removes the bias, because the model was not refined against $\mathcal{F}$:

$$(\hat{D}_{hk}^{\text{LB}}, \hat{\Sigma}_{hk}^{\text{LB}}) = \arg\max_{D,\Sigma} \prod_{\mathbf{h}\in\mathcal{F}_k} P(|F_{\text{obs}}(\mathbf{h})| \mid |F_{\text{mod}}(\mathbf{h})|, D, \Sigma) \tag{42}$$

The resulting phase probability is:

$$P_{\text{LB}}(\varphi_h) \propto \exp\left[\alpha_k\, |F_{\text{obs}}||F_{\text{mod}}| \cos(\varphi_h - \varphi_{\text{mod}}) + \beta_k\, |F_{\text{obs}}|^2\right] \tag{43}$$

where $\alpha_k = 2\hat{D}_k^{\text{LB}}/\hat{\Sigma}_{hk}^{\text{LB}}$ and $\beta_k = -1/\hat{\Sigma}_{hk}^{\text{LB}}$.

### 6.2 Equivalence to von Mises

**Lemma 6.1. LB distribution is von Mises**

The LB phase distribution (43) is von Mises with:

$$\varphi_{\text{mod}}^{\text{LB}} = \varphi_{\text{mod}} \quad \text{(same phase centre)} \tag{44}$$

$$X_h^{\text{LB}} = \alpha_k\, |F_{\text{obs}}||F_{\text{mod}}| \tag{45}$$

The term $\beta_k|F_{\text{obs}}|^2$ is phase-independent and contributes only to the normalisation constant.

*Proof.* In (43), only $\cos(\varphi_h - \varphi_{\text{mod}})$ depends on $\varphi_h$. Therefore $P_{\text{LB}}(\varphi_h) \propto \exp[\alpha_k|F_{\text{obs}}||F_{\text{mod}}| \cos(\varphi_h - \varphi_{\text{mod}})]$, which is the von Mises kernel with $\mu = \varphi_{\text{mod}}$ and $\kappa = \alpha_k|F_{\text{obs}}||F_{\text{mod}}|$. □ □

### 6.3 Connection between all three model-based maps

**Theorem 6.1. Nesting of model-based estimators**

The three model-based maps satisfy the following inequalities in MSE and the following ordering in FOM space:

$$m_h^{\text{true}} \le m_h^{\text{LB}} \approx m_h^{\text{true}} \le m_h^{\Sigma A} \tag{46}$$

and correspondingly for the maps:

$$\text{MSE}[\rho_{\text{BC}}] \le \text{MSE}[\rho_{\text{LB}}] \le \text{MSE}[\rho_{\Sigma\text{A}}] \tag{47}$$

*assuming the model has been refined against the working set.*

*Proof.* LB estimates are consistent (converge to $m^{\text{true}}$ as $|\mathcal{F}| \to \infty$) by the consistency of ML estimation on the uncontaminated R-free set. Sigma-A estimates are inconsistent (biased upward) as shown in Proposition 5.3. The MSE ordering follows because overestimated FOMs in $\rho_{\Sigma\text{A}}$ amplify noise at each reflection, and underestimated FOMs in $\rho_{\text{BC}}$ (if the true $m_h$ were known exactly) would be optimal. □ □

The practical implication: REFMAC's standard $2F_o - F_c$ map has larger MSE than the LB-estimated map, which has larger MSE than the theoretical Blow-Crick optimum. None of these is achievable without additional information.

# §7 Composite OMIT Maps

## 7.1 The composite OMIT operation as a projection

Composite OMIT maps (Terwilliger et al. 2008) partition the unit cell into $M$ overlapping sub-volumes $\{V_j\}_{j=1}^{M}$ and compute:

$$\rho_j^{\mathrm{OMIT}}(\mathbf{r}) = \mathcal{F}^{-1}\big[|F_{\mathrm{obs}}|\, e^{i\varphi_j^{\mathrm{omit}}}\big](\mathbf{r}), \quad \mathbf{r} \in V_j \tag{48}$$

where $\varphi_j^{\mathrm{omit}}$ is computed from a model with all atoms in $V_j$ omitted (via simulated annealing refinement). The composite map is:

$$\rho^{\mathrm{OMIT}}(\mathbf{r}) = \sum_{j=1}^{M} \chi_j(\mathbf{r})\rho_j^{\mathrm{OMIT}}(\mathbf{r}) \tag{49}$$

where $\chi_j$ is the indicator function of $V_j$.

## 7.2 Lack of a closed-form weight

The composite OMIT map **cannot be written in the form** (21) with a single pair $(w_h, \hat{\varphi}_h)$ per reflection, because: (i) each sub-volume uses different phases $\varphi_j^{\mathrm{omit}}$, producing a spatially heterogeneous Fourier synthesis; and (ii) the tiling (49) mixes contributions from different phase sets. In Fourier space:

$$\mathcal{F}[\rho^{\mathrm{OMIT}}](\mathbf{h}) = \sum_j \mathcal{F}[\chi_j \cdot \rho_j^{\mathrm{OMIT}}](\mathbf{h}) = \sum_j \big(\hat{\chi}_j * (|F_{\mathrm{obs}}|\, e^{i\varphi_j^{\mathrm{omit}}})\big)(\mathbf{h}) \tag{50}$$

where $*$ denotes convolution. This is a convolution of the phase field with the sub-volume shape functions, not a simple per-reflection weighting. Consequently, composite OMIT maps have no formal connection to the Blow-Crick minimum MSE theory, no FOM estimates, and no statistically characterised uncertainty.

# §8 PanDDA: Amplitude Statistics Across Datasets

## 8.1 The PanDDA ground-state map

PanDDA operates in a fundamentally different paradigm: it does not use model phases at all for ground-state estimation. Instead, for $N$ datasets the ground-state amplitude per reflection is:

$$|\bar{F}(\mathbf{h})| = \frac{1}{N}\sum_{d=1}^{N} s_d\, |F_{\mathrm{obs}}^{(d)}(\mathbf{h})| \tag{51}$$

The ground-state *map* requires phases, which PanDDA obtains from a single refined model $\varphi_{\mathrm{mod}}$:

$$\rho^{\mathrm{GS}}(\mathbf{r}) = \mathcal{F}^{-1}\big[|\bar{F}|\, e^{i\varphi_{\mathrm{mod}}}\big](\mathbf{r}) \tag{52}$$

In the limit $N \to \infty$ with most datasets being apo: $|\bar{F}(\mathbf{h})| \to |F_{\mathrm{true}}^{\mathrm{apo}}(\mathbf{h})|$, so the ground state approaches the true apo density.

## 8.2 The event map as difference density

For dataset $d$ with a bound ligand at occupancy $q$:

$$F_{\mathrm{obs}}^{(d)}(\mathbf{h}) = (1-q)\, F_{\mathrm{true}}^{\mathrm{apo}}(\mathbf{h}) + q\, F_{\mathrm{true}}^{\mathrm{bound}}(\mathbf{h}) + \mathrm{noise} \tag{53}$$

The BDC-corrected event map is:

$$\rho_d^{\text{event}}(\mathbf{r}) = \mathcal{F}^{-1}\big[|F_{\text{obs}}^{(d)}|e^{i\varphi_{\text{mod}}}\big](\mathbf{r}) - (1-\alpha)\,\rho^{\text{GS}}(\mathbf{r}) \tag{54}$$

where $\alpha = \text{BDC} \in [0,1]$. Writing in Fourier space:

$$\mathcal{F}[\rho_d^{\text{event}}](\mathbf{h}) = \big(|F_{\text{obs}}^{(d)}| - (1-\alpha)|\bar{F}|\big)\, e^{i\varphi_{\text{mod}}} \tag{55}$$

The effective amplitude weight is 1 (no FOM) but the amplitude itself is the difference $|F_{\text{obs}}^{(d)}| - (1-\alpha)|\bar{F}|$.

### 8.3 PanDDA SNR scaling

The signal at the ligand site in the event map:

$$\rho_d^{\text{event}}\big|_{\text{site}} \approx q \cdot \rho^{\text{bound}}\big|_{\text{site}} + \mathcal{O}(1/\sqrt{N}) \tag{56}$$

The noise term $\mathcal{O}(1/\sqrt{N})$ is the sampling error in $|\bar{F}|$ from finite $N$. The key result: the signal scales as $q$, the noise as $N^{-1/2}$, so the SNR scales as $q\sqrt{N}$.

## §9 The EDENN Stochastic Framework

### 9.1 The stochastic omit map

For each iteration $k = 1, \ldots, K$, a subset $\mathcal{O}_k$ of atoms is randomly omitted (fraction $P_{\text{omit}}$), yielding phases:

$$\varphi^{(k)}(\mathbf{h}) = \arg\left(\sum_{j \notin \mathcal{O}_k} f_j(s)\, e^{2\pi i\, \mathbf{h}\cdot\mathbf{r}_j}\right) \tag{57}$$

The $k$-th omit map uses these phases with observed amplitudes:

$$\rho^{(k)}(\mathbf{r}) = \mathcal{F}^{-1}\big[|F_{\text{obs}}|\, e^{i\varphi^{(k)}}\big](\mathbf{r}) \tag{58}$$

### 9.2 The ensemble mean as Blow-Crick map: full derivation

**Theorem 9.1. EDENN mean map identity**

In the limit $K \to \infty$, the ensemble mean of the omit maps converges in $L^2(V)$ to the Blow-Crick best map with FOM determined by the phase perturbation distribution induced by the omit protocol:

$$\lim_{K\to\infty} \frac{1}{K}\sum_{k=1}^{K} \rho^{(k)}(\mathbf{r}) = \rho_{\text{BC}}(\mathbf{r}) = \mathcal{F}^{-1}\big[m_h^{\text{EDENN}}\, |F_{\text{obs}}|\, e^{i\varphi_{\text{mod}}}\big](\mathbf{r}) \tag{59}$$

where $m_h^{\text{EDENN}} = \mathbb{E}_{\mathcal{O}}[\cos(\varphi^{(k)}(\mathbf{h}) - \varphi_{\text{mod}}(\mathbf{h}))]$.

*Proof.* By the strong law of large numbers applied in $L^2(V)$ (which is a separable Hilbert space and the maps $\rho^{(k)}$ are bounded):

$$\frac{1}{K}\sum_{k=1}^{K} \rho^{(k)} \xrightarrow{L^2} \mathbb{E}_{\mathcal{O}}[\rho^{(k)}] \tag{60}$$

almost surely. The expectation is:

$$\begin{aligned}\mathbb{E}_{\mathcal{O}}[\rho^{(k)}(\mathbf{r})] &= \mathbb{E}_{\mathcal{O}}\left[\mathcal{F}^{-1}[|F_{\text{obs}}|e^{i\varphi^{(k)}}](\mathbf{r})\right] \\ &= \mathcal{F}^{-1}\left[\mathbb{E}_{\mathcal{O}}\left[|F_{\text{obs}}(\mathbf{h})|\, e^{i\varphi^{(k)}(\mathbf{h})}\right]\right](\mathbf{r})\end{aligned} \tag{61}$$

where we interchange expectation and $\mathcal{F}^{-1}$ by dominated convergence (the maps are uniformly bounded in $L^2$). Now:

$$\mathbb{E}_{\mathcal{O}}\left[|F_{\text{obs}}(\mathbf{h})|\, e^{i\varphi^{(k)}(\mathbf{h})}\right] = |F_{\text{obs}}(\mathbf{h})|\, \mathbb{E}_{\mathcal{O}}\left[e^{i\varphi^{(k)}(\mathbf{h})}\right] \tag{62}$$

The random variable $\varphi^{(k)}(\mathbf{h})$ is distributed approximately as vonMises$(\varphi_{\text{mod}}(\mathbf{h}), X_h)$ for large models (by the central limit theorem for circular statistics applied to the sum of random complex exponentials). The characteristic function of vonMises$(\mu, \kappa)$ at argument $n = 1$ is:

$$\mathbb{E}[e^{i\varphi}] = \frac{I_1(\kappa)}{I_0(\kappa)}\, e^{i\mu} = m_h\, e^{i\varphi_{\text{mod}}(\mathbf{h})} \tag{63}$$

Substituting (63) into (62) and then into (61):

$$\mathbb{E}_{\mathcal{O}}[\rho^{(k)}(\mathbf{r})] = \mathcal{F}^{-1}\big[m_h\, |F_{\text{obs}}|\, e^{i\varphi_{\text{mod}}}\big](\mathbf{r}) = \rho_{\text{BC}}(\mathbf{r}) \tag{64}$$

This completes the proof. □ □

## 9.3 Justification of the von Mises approximation

**Lemma 9.1. CLT for omit phases**

Let $F^{(k)}_{\text{omit}}(\mathbf{h}) = \sum_{j \notin \mathcal{O}_k} f_j(s) e^{2\pi i \mathbf{h}\cdot\mathbf{r}_j}$. For large $N_{\text{atoms}}$ with bounded scattering factors, the fluctuation $F^{(k)}_{\text{omit}}(\mathbf{h}) - \mathbb{E}[F^{(k)}_{\text{omit}}(\mathbf{h})]$ converges in distribution to a bivariate normal:

$$\sqrt{N}\big(F^{(k)}_{\text{omit}} - (1 - P_{\text{omit}})F_{\text{mod}}\big) \xrightarrow{d} \mathcal{N}_{\mathbb{C}}(0, \Sigma_h) \tag{65}$$

where $\Sigma_h = P_{\text{omit}}(1 - P_{\text{omit}}) \sum_j |f_j(s)|^2$.

*Proof.* Apply the Lyapunov CLT to the sum of independent (across $j$) complex random variables $Z^{(k)}_j = \mathbf{1}[j \notin \mathcal{O}_k] \cdot f_j(s) e^{2\pi i \mathbf{h}\cdot\mathbf{r}_j}$. The Lyapunov condition is satisfied because $|f_j(s)|$ is bounded. □ □

For a complex Gaussian perturbation around $(1 - P_{\text{omit}})F_{\text{mod}}$, the resulting distribution of the phase $\varphi^{(k)}(\mathbf{h})$ is approximately von Mises with concentration:

$$X^{\text{EDENN}}_h \approx \frac{2(1 - P_{\text{omit}})^2 |F_{\text{mod}}|^2}{\Sigma_h} = \frac{2(1 - P_{\text{omit}})}{P_{\text{omit}}} \frac{|F_{\text{mod}}|^2}{\sum_j |f_j|^2} \tag{66}$$

## 9.4 The FOM estimator from Fourier analysis

**Proposition 9.1. Reciprocal-space FOM estimator**

The per-reflection FOM can be estimated from the ensemble as:

$$\hat{m}^{\text{EDENN}}_h = \frac{|\mathcal{F}[\langle \rho^{\text{omit}} \rangle](\mathbf{h})|}{|F_{\text{obs}}(\mathbf{h})|} = \frac{|\langle \tilde{F}^{\text{omit}}(\mathbf{h}) \rangle|}{|F_{\text{obs}}(\mathbf{h})|} \tag{67}$$

This estimator is unbiased in the limit $K \to \infty$: $\mathbb{E}[\hat{m}_h^{\mathrm{EDENN}}] = m_h^{\mathrm{EDENN}}$.

*Proof.* From (64): $\mathcal{F}[\langle\rho^{\mathrm{omit}}\rangle](\mathbf{h}) = m_h|F_{\mathrm{obs}}|e^{i\varphi_{\mathrm{mod}}}$. Taking moduli and dividing by $|F_{\mathrm{obs}}|$ gives (67). The finite-$K$ estimator $\hat{m}_h^K = |\frac{1}{K}\sum_k e^{i\varphi^{(k)}}|$ satisfies $\mathbb{E}[\hat{m}_h^K] = m_h^{\mathrm{EDENN}}$ by linearity of expectation and the law of large numbers. □ □

### 9.5 The real-space SNR and its connection to FOM

The SNR map is:

$$\mathrm{SNR}(\mathbf{r}) = \frac{\langle\rho^{\mathrm{omit}}\rangle(\mathbf{r})}{\mathrm{std}[\rho^{(k)}(\mathbf{r})]} \tag{68}$$

In the isolated-site approximation (a single reflection $\mathbf{h}^*$ dominates the density at $\mathbf{r}^*$):

**Proposition 9.2. SNR–FOM conversion**

At a site dominated by reflection $\mathbf{h}^*$:

$$\mathrm{SNR}(\mathbf{r}^*) \approx m_h^* \cdot \sqrt{\frac{2}{1-(m_h^*)^2}} \tag{69}$$

Inverting:

$$m_h^* \approx \frac{\mathrm{SNR}}{\sqrt{2+\mathrm{SNR}^2}} \tag{70}$$

*Proof.* At position $\mathbf{r}^*$, $\rho^{(k)}(\mathbf{r}^*) \approx |F_{\mathrm{obs}}(\mathbf{h}^*)|\cos(\varphi^{(k)} - 2\pi\mathbf{h}^*\cdot\mathbf{r}^*)$. For the von Mises distribution of $\varphi^{(k)}$: $\mathbb{E}[\cos(\varphi^{(k)}-\phi^*)] = m_h\cos(\varphi_{\mathrm{mod}}-\phi^*)$, $\mathrm{Var}[\cos(\varphi^{(k)}-\phi^*)] = \frac{1}{2}(1-m_h^2)$ (using the von Mises second moment $\mathbb{E}[\cos^2\theta] = \frac{1}{2}(1+I_2/I_0)$ and $I_2/I_0 \approx m_h^2$ for moderate $X_h$). Therefore $\mathrm{SNR}^2 = m_h^2/(\frac{1}{2}(1-m_h^2)) = 2m_h^2/(1-m_h^2)$. Solving for $m_h$ gives (70). □ □

## §10 The Difference Density and Low-Occupancy Detection

### 10.1 Construction

The stochastic difference density is:

$$\Delta\rho(\mathbf{r}) = \rho_{\mathrm{obs}}(\mathbf{r}) - \langle\rho^{\mathrm{omit}}\rangle(\mathbf{r}) = \mathcal{F}^{-1}\big[|F_{\mathrm{obs}}|\,e^{i\varphi_{\mathrm{obs}}} - m_h|F_{\mathrm{obs}}|\,e^{i\varphi_{\mathrm{mod}}}\big](\mathbf{r}) \tag{71}$$

In Fourier space:

$$\mathcal{F}[\Delta\rho](\mathbf{h}) = |F_{\mathrm{obs}}(\mathbf{h})|\,e^{i\varphi_{\mathrm{obs}}(\mathbf{h})}\big(1 - m_h\,e^{i(\varphi_{\mathrm{mod}}-\varphi_{\mathrm{obs}})}\big) \tag{72}$$

### 10.2 Low-occupancy signal analysis

For a ligand at occupancy $q$ absent from the model:

$$F_{\mathrm{obs}}(\mathbf{h}) = F^{\mathrm{prot}}(\mathbf{h}) + qF^{\mathrm{lig}}(\mathbf{h}) + \mathrm{noise} \tag{73}$$

$$\varphi_{\mathrm{obs}}(\mathbf{h}) = \varphi_{\mathrm{mod}}(\mathbf{h}) + \delta_h + \mathcal{O}(q^2) \tag{74}$$

where $\delta_h = q\,\mathrm{Im}[F^{\mathrm{lig}}(\mathbf{h})/F^{\mathrm{prot}}(\mathbf{h})]$ is the phase shift due to the ligand.

**Theorem 10.1. Difference density linearity in occupancy**

To first order in $q$:

$$\mathcal{F}[\Delta\rho](\mathbf{h}) \approx q\,F^{\mathrm{lig}}(\mathbf{h}) + \mathcal{O}(q^2) \tag{75}$$

Therefore:

$$\Delta\rho(\mathbf{r}) \approx q\,\rho^{\text{lig}}_{\text{full}}(\mathbf{r}) + \text{noise} \tag{76}$$

where $\rho^{\text{lig}}_{\text{full}}$ is the full-occupancy ligand density. The SNR at the ligand site scales as:

$$\text{SNR}[\Delta\rho]\big|_{\text{site}} \approx \frac{q\,|F^{\text{lig}}(\mathbf{h}^*)|}{\sigma_{\text{ens}}} \tag{77}$$

with **no FOM attenuation factor**.

*Proof.* Expand to first order in $q$:

$$|F_{\text{obs}}(\mathbf{h})| = |F^{\text{prot}}| + q\,\text{Re}[F^{\text{lig}}e^{-i\varphi_{\text{mod}}}] + \mathcal{O}(q^2) \tag{78}$$

$$\varphi_{\text{obs}}(\mathbf{h}) = \varphi_{\text{mod}} + q\,\frac{\text{Im}[F^{\text{lig}}e^{-i\varphi_{\text{mod}}}]}{|F^{\text{prot}}|} + \mathcal{O}(q^2) \tag{79}$$

Substituting into $|F_{\text{obs}}|e^{i\varphi_{\text{obs}}}$ and expanding:

$$|F_{\text{obs}}|e^{i\varphi_{\text{obs}}} = |F^{\text{prot}}|e^{i\varphi_{\text{mod}}} + qF^{\text{lig}} + \mathcal{O}(q^2) \tag{80}$$

The ensemble mean gives $m_h|F_{\text{obs}}|e^{i\varphi_{\text{mod}}}$. Since $m_h \approx 1$ for the well-modelled protein and $|F_{\text{obs}}| \approx |F^{\text{prot}}|$ to leading order:

$$|F_{\text{obs}}|e^{i\varphi_{\text{obs}}} - m_h|F_{\text{obs}}|e^{i\varphi_{\text{mod}}} \approx (1-m_h)|F^{\text{prot}}|e^{i\varphi_{\text{mod}}} + qF^{\text{lig}} \tag{81}$$

The term $(1-m_h)|F^{\text{prot}}|e^{i\varphi_{\text{mod}}}$ is the residual ensemble noise (uncorrelated with the ligand signal for random omit selections), contributing to $\sigma_{\text{ens}}$ but not to the ligand signal. Therefore the ligand contribution to $\mathcal{F}[\Delta\rho]$ is $qF^{\text{lig}}$ to first order, establishing (75). □ □

### 10.3 Comparison with mean-map sensitivity

The mean-map signal at the ligand site:

$$\langle\rho^{\text{omit}}\rangle\big|_{\text{site}} \approx m_h \cdot \cos(\delta_h) \cdot q \cdot \rho^{\text{lig}}_{\text{full}}\big|_{\text{site}} \tag{82}$$

For $m_h = 0.7$, $\delta_h = 20$, $q = 0.10$: mean-map attenuation factor $= 0.7 \times \cos(20) \times 0.1 \approx 0.066$. Difference-density attenuation factor $= 0.1$ (factor of $1.5\times$ improvement at this occupancy; grows to $\gg 10\times$ as $q \to 0$).

## §11 Master Connectivity Diagram

### 11.1 All maps as functionals of the true density

Every map discussed can be written as a linear operator applied to the true density $\rho_{\text{true}}$ plus a noise/bias term. Table 1 summarises the exact relationships.

| **Map** | Fourier representation | Relation to $\rho_{\text{true}}$ |
|---|---|---|
| **Ground truth** $\rho_{\text{true}}$ | $\mathcal{F}^{-1}[\lvert F_{\text{true}}\rvert e^{i\varphi_{\text{true}}}]$ | — |
| **SAD-phased** $\rho_{\text{SAD}}$ | $\mathcal{F}^{-1}[m_h^{\text{SAD}}\lvert F_{\text{obs}}\rvert e^{i\varphi^{\text{SAD}}_{\text{mod}}}]$ | $= \mathbb{E}_{P_{\text{SAD}}}[\rho_{\text{true}}] + \delta_{\text{amp}}$ |
| **Blow-Crick** $\rho_{\text{BC}}$ | $\mathcal{F}^{-1}[m_h\lvert F_{\text{obs}}\rvert e^{i\varphi_{\text{mod}}}]$ | $= \mathbb{E}_{P(\varphi_h)}[\rho_{\text{true}}] + \delta_{\text{amp}}$ |

| **Map** | Fourier representation | Relation to $\rho_{\text{true}}$ |
|---|---|---|
| **Sigma-A** $\rho_{\Sigma\text{A}}$ | $\mathcal{F}^{-1}[m_h^{\Sigma A}\lvert F_{\text{obs}}\rvert e^{i\varphi_{\text{mod}}}]$ | $= \rho_{\text{BC}} + \mathcal{F}^{-1}[(m_h^{\Sigma A} - m_h)\lvert F_{\text{obs}}\rvert e^{i\varphi_{\text{mod}}}]$ |
| **Lunin LB** $\rho_{\text{LB}}$ | $\mathcal{F}^{-1}[m_h^{\text{LB}}\lvert F_{\text{obs}}\rvert e^{i\varphi_{\text{mod}}}]$ | $\approx \rho_{\text{BC}}$ (unbiased) |
| **EDENN mean** $\langle\rho^{\text{omit}}\rangle$ | $\mathcal{F}^{-1}[m_h^{\text{EDENN}}\lvert F_{\text{obs}}\rvert e^{i\varphi_{\text{mod}}}]$ | $= \rho_{\text{BC}}$ (as $K \to \infty$) |
| **EDENN diff.** $\Delta\rho$ | $\mathcal{F}^{-1}[(1 - m_h)\lvert F_{\text{obs}}\rvert e^{i\varphi_{\text{mod}}} + qF^{\text{lig}}]$ | $\approx q\rho_{\text{full}}^{\text{lig}} + \text{noise}$ |
| **PanDDA event** | $\mathcal{F}^{-1}[(\lvert F_{\text{obs}}^{(d)}\rvert - (1-\alpha)\lvert\bar{F}\rvert)e^{i\varphi_{\text{mod}}}]$ | $\approx q\rho_{\text{full}}^{\text{lig}} + \mathcal{O}(N^{-1/2})$ |
| **Composite OMIT** | See eq. (50) | No closed form; not MMSE |

Table 1: Fourier representations and relationships to the true density. $\delta_{\text{amp}} = \mathcal{F}^{-1}[(\lvert F_{\text{obs}}\rvert - \lvert F_{\text{true}}\rvert) \cdot w_h e^{i\hat{\varphi}}]$ is the amplitude-noise correction (small).

### 11.2 The commuting operator diagram

Define the following operators:

$$\mathcal{A}_w : F(\mathbf{h}) \mapsto w_h F(\mathbf{h}) \quad \text{(amplitude weighting)} \tag{83}$$

$$\mathcal{P}_\mu : F(\mathbf{h}) \mapsto \lvert F(\mathbf{h})\rvert e^{i\mu(\mathbf{h})} \quad \text{(phase replacement)} \tag{84}$$

$$\mathcal{E}_K : \{F^{(k)}\} \mapsto \frac{1}{K}\sum_k F^{(k)} \quad \text{(ensemble average)} \tag{85}$$

Then the connectivity can be expressed as:

$$\rho_{\text{BC}} = \mathcal{F}^{-1} \circ \mathcal{A}_{m_h} \circ \mathcal{P}_{\varphi_{\text{mod}}}[\lvert F_{\text{obs}}\rvert] \tag{86}$$

$$\langle\rho^{\text{omit}}\rangle = \mathcal{F}^{-1} \circ \mathcal{E}_K \circ \mathcal{P}_{\{\varphi^{(k)}\}}[\lvert F_{\text{obs}}\rvert] \tag{87}$$

$$\langle\rho^{\text{omit}}\rangle \xrightarrow{K\to\infty} \rho_{\text{BC}} \tag{88}$$

The convergence (88) is the central result: $\mathcal{E}_\infty \circ \mathcal{P}_{\{\varphi^{(k)}\}} = \mathcal{A}_{m_h} \circ \mathcal{P}_{\varphi_{\text{mod}}}$ — ensemble averaging over stochastic phases is equivalent to FOM weighting the model phase.

### 11.3 The FOM chain

All FOM estimators target the same quantity $m_h^{\text{true}}$:

$$m_h^{\text{true}} = \mathbb{E}_{\varphi_h}[\cos(\varphi_h - \varphi_{\text{mod}})] \tag{89}$$

Their relationships form a strict ordering:

$$0 \le m_h^{\text{true}} \underset{(1)}{\le} m_h^{\text{EDENN}}, m_h^{\text{LB}} \underset{(2)}{\le} m_h^{\Sigma A} \underset{(3)}{\le} 1 \tag{90}$$

where inequality (1) reflects residual bias from finite $K$ and imperfect von Mises approximation; inequality (2) is the working-set inflation; and inequality (3) is the definition of FOM.

## §12 Experimental Phases as Ground Truth: Formal Status

### 12.1 The epistemic hierarchy

The following quantities are strictly ordered by their proximity to $\varphi_{\text{true}}(\mathbf{h})$:

1. $\varphi_{\text{true}}(\mathbf{h})$: exists mathematically per equation (4), but is **never directly observable**.
2. $\varphi_{\text{mod}}^{\text{SAD}}(\mathbf{h})$: posterior mean from anomalous dispersion. Best experimentally accessible estimate. Residual error $\sim$ 20–40 at 2 Å.
3. $\varphi_{\text{mod}}^{\text{MAD}}(\mathbf{h})$: sharpened posterior from multi-wavelength anomalous dispersion. Residual error $\sim$ 15–25 at 2 Å.
4. $\varphi_{\text{mod}}^{\text{model}}(\mathbf{h})$: phase from refined atomic model. Accuracy depends on model quality. Residual error $\sim$ 30–60 for typical structures.

### 12.2 Implication for EDENN validation

The EDENN FOM estimator $\hat{m}_h^{\text{EDENN}}$ can be validated by treating SAD phases as approximate ground truth:

$$m_h^{\text{emp}}(\mathbf{h}) = \cos\big(\varphi_{\text{mod}}(\mathbf{h}) - \varphi_{\text{mod}}^{\text{SAD}}(\mathbf{h})\big) \tag{91}$$

The expected relationship between $\hat{m}_h^{\text{EDENN}}$ and $m_h^{\text{emp}}$ (the Lunin & Skovoroda Figure 1 equivalent) is:

$$\mathbb{E}[\hat{m}_h^{\text{EDENN}}] \approx m_h^{\text{emp}} + \mathcal{O}(\sigma_{\text{SAD}}^2) \tag{92}$$

where $\sigma_{\text{SAD}}^2$ is the residual variance of SAD phases relative to the true phase. This correction is second-order in the SAD phase error, so for well-determined SAD phases it is small.

## §13 Summary: What Each Map Tells You and What It Cannot

### 13.1 Formal properties

Every map in the family (21) has:

- **Optimal FOM** $\iff$ $w_h = m_h^{\text{true}}$ (Blow-Crick theorem)
- **Unbiased FOM** $\iff$ parameters estimated from held-out data (LB estimates, EDENN)
- **Full phase posterior** $\iff$ von Mises distribution with calibrated $X_h$ (EDENN only)
- **Spatial uncertainty** $\iff$ per-voxel SNR (EDENN only)
- **Low-occupancy sensitivity** $\iff$ difference density or amplitude subtraction (EDENN difference, PanDDA)

### 13.2 The strict containment

The EDENN framework strictly contains every other paradigm as a special case:

**Corollary 13.1. Containment**

(i) As $K \to \infty$ with optimal FOM: EDENN $\to$ Blow-Crick

(ii) As $K \to \infty$ with working-set phase estimation: EDENN $\to$ Sigma-A map

(iii) As $K \to \infty$ with R-free phase estimation: EDENN $\to$ LB map

(iv) With SAD phases as $\varphi_{\text{mod}}$: EDENN mean map $\to$ SAD-phased map

(v) Replacing ensemble mean with dataset mean: EDENN → PanDDA ground state

*Proof.* (i)–(iv) follow directly from Theorem 9.1 with different choices of $\varphi_{\text{mod}}$ and FOM estimation procedure. (v) holds because $\frac{1}{K}\sum_k |F_{\text{obs}}^{(k)}| e^{i\varphi_{\text{mod}}}$ (amplitudes averaged over datasets, fixed phase) equals the PanDDA ground state (52) when the $K$ omit iterations correspond to $N$ different datasets. □ □

### 13.3 Concluding statement

The ground-truth map $\rho_{\text{true}}$ is the unique element of $L^2(V)$ corresponding to the complete, noise-free, true atomic structure. It is never directly accessible.

All crystallographic maps are *estimators* of $\rho_{\text{true}}$, defined by their choice of $(w_h, \hat{\varphi}_h)$ and characterised by their MSE (23). The hierarchy of estimators, from most to least biased and from least to most informative, is:

$$\text{standard 2Fo-Fc} \succ \text{Sigma-A} \succ \text{LB} \approx \text{EDENN mean} \succ \text{Blow-Crick} \succ \text{true density} \tag{93}$$

where $A \succ B$ means $\text{MSE}[A] > \text{MSE}[B]$.

The EDENN framework's unique contribution is to provide, simultaneously: the MMSE mean estimate, the full phase probability distribution, the real-space uncertainty map, and the difference density for low-occupancy detection — all from a single stochastic algorithm with a formally characterised convergence rate of $O(K^{-1/2})$.

## §A Von Mises Distribution: Key Properties

For $\varphi_h \sim \text{vonMises}(\mu, \kappa)$:

$$P(\varphi_h) = \frac{e^{\kappa\cos(\varphi_h - \mu)}}{2\pi I_0(\kappa)} \tag{94}$$

$$\mathbb{E}[e^{in\varphi_h}] = \frac{I_{|n|}(\kappa)}{I_0(\kappa)} e^{in\mu}, \quad n \in \mathbb{Z} \tag{95}$$

$$\mathbb{E}[\cos(\varphi_h - \mu)] = \frac{I_1(\kappa)}{I_0(\kappa)} \equiv m_h \tag{96}$$

$$\text{Var}[\cos(\varphi_h - \mu)] = \tfrac{1}{2}\left(1 + \frac{I_2(\kappa)}{I_0(\kappa)}\right) - m_h^2 \tag{97}$$

$$H[\varphi_h] = \ln(2\pi I_0(\kappa)) - \kappa m_h \quad \text{(differential entropy)} \tag{98}$$

$$\kappa \to 0 \implies P(\varphi_h) \to \text{Uniform}(-\pi, \pi), \quad m_h \to 0 \tag{99}$$

$$\kappa \to \infty \implies P(\varphi_h) \to \delta(\varphi_h - \mu), \quad m_h \to 1 \tag{100}$$

## §B Centric Reflections

Centric reflections satisfy $\varphi_h \in \{0, \pi\}$ by crystal symmetry. The phase distribution is Bernoulli:

$$P(\varphi_h = \varphi_{\text{mod}}) = \frac{e^{\kappa/2}}{2\cosh(\kappa/2)} \tag{101}$$

$$P(\varphi_h = \varphi_{\text{mod}} + \pi) = \frac{e^{-\kappa/2}}{2\cosh(\kappa/2)} \tag{102}$$

$$m_h^{\text{centric}} = \tanh(\kappa/2) \tag{103}$$

All theorems above apply to centric reflections with $I_1/I_0$ replaced by $\tanh(\kappa/2)$.

## §C Parseval's Theorem and Map Comparison

For two maps $\rho^A$ and $\rho^B$ in $L^2(V)$:

$$\int_V |\rho^A(\mathbf{r}) - \rho^B(\mathbf{r})|^2 \, d\mathbf{r} = \frac{1}{V} \sum_{\mathbf{h}} |\hat{\rho}^A(\mathbf{h}) - \hat{\rho}^B(\mathbf{h})|^2 \tag{104}$$

This identity, used in the proof of Theorem 4.1, reduces all $L^2$ map comparisons to per-reflection squared differences in reciprocal space, making the optimisation separable.

## §C Key References

**Blow & Crick (1959)**
Derivation of the best Fourier map. *Acta Cryst.* **12**, 794.

**Hendrickson (1979)**
Phase information from anomalous dispersion of synchrotron radiation. *Acta Cryst. A* **35**, 245.

**Read (1986)**
Improved Fourier coefficients using Sigma-A. *Acta Cryst. A* **42**, 140.

**Brünger (1992)**
Free R-value. *Nature* **355**, 472.

**Lunin & Skovoroda (1995)**
R-free LB estimates. *Acta Cryst. A* **51**, 880.

**Murshudov et al. (1997)**
REFMAC5 refinement. *Acta Cryst. D* **53**, 240.

**Terwilliger et al. (2008)**
Composite OMIT maps in Phenix. *Acta Cryst. D* **64**, 61.

**Liebschner et al. (2017)**
Polder maps. *Acta Cryst. D* **73**, 148.

**Pearce et al. (2017)**
PanDDA. *Nat. Commun.* **8**, 15816.